An

INTERNSHIP REPORT

on

DATA ANALYTICS WITH TABLEAU

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING

by

NARASIMHA KAIPU
(Roll No. 23AK5A0514)

Under the mentorship of
Mr .M.Poojitha, MCA,
Assistant Professor, CSE Department

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ANNAMACHARYA INSTITUTE OF TECHNOLOGY AND SCIENCES, TIRUPATI
(AUTONOMOUS)

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that Ms**. NARASIMHA KAIPU(23AK5A0514)** has carried out Virtual Internship on "**DATA ANALYTICS WITH TABLEAU**" and submitted to the Department of **COMPUTER SCIENCE AND ENGINEERING** of Annamacharya Institute of Technology and Sciences, in partial fulfilment of the requirements for the award of the Degree of **Bachelor of Technology** in Computer Science and Engineering is meeting the Academic Regulations.

Signature of Internship Mentor

Signature of HOD

DECLARATION

I am **NARASIMHA KAIPU(23AK5A0514),** Studying Final year B.Tech in Computer Science and Engineering of Annamacharya Institute of Technology and Sciences, hereby declare that this Internship report titled "**DATA ANALYTICS WITH TABLEAU**" has been done by me. The Internship work carried out is original and has not been submitted to any other University or Institution for the award of any credits. I promise to meet all the mandatory requirements as specified by the Academic regulations.

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ACKNOWLEDGEMENT

The needs and deeds of a particular person are only satisfied with the support and endurance of many.

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NARASIMHA KAIPU

Roll NO: 23AK5A0514

ABSTRACT

This project, "**PLUGGING INTO THE FUTURE: AN EXPLORATION OF ELECTRICITY CONSUMPTION**" is a data visualization project that focuses on analyzing electricity consumption across different regions, sectors, and time periods. The project utilizes Tableau to transform raw electricity usage data into interactive and meaningful visual insights.

The study examines key factors such as time-of-day usage, seasonal variations, peak demand periods, and sector-wise consumption including residential, commercial, and industrial sectors. Special emphasis is given to understanding how electricity demand changes across regions and how external conditions influence consumption patterns.

Using Tableau features like dashboards, filters, calculated fields, and forecasting, the project presents clear visualizations that help identify trends, high-demand areas, and usage behavior. The analysis enables better understanding of electricity distribution and highlights opportunities for optimizing energy usage.

The results of this project provide valuable insights for utility companies, policymakers, and consumers by supporting efficient energy management, improving grid planning, and encouraging sustainable electricity practices. This project demonstrates how data visualization tools can play a crucial role in decision-making and future energy planning.

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   Verifiable Credentials

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CHAPTER- 1

INTRODUCTION

CHAPTER-1
INTRODUCTION

INTRODUCTION

The project titled "**Plugging into the Future: An Exploration of Electricity Consumption**" focuses on analyzing electricity usage patterns to generate meaningful and actionable insights for stakeholders such as policymakers, energy providers, researchers, and consumers. Electricity plays a fundamental role in modern society, acting as the backbone of economic development, technological advancement, and improved quality of life. From powering industries and transportation systems to supporting daily household activities and digital communication, electricity is an essential resource. With the rapid growth in population, urbanization, and industrialization, the demand for electricity has increased significantly, making it crucial to understand consumption patterns and optimize energy usage.

In today's digital era, vast amounts of electricity consumption data are generated from multiple sources such as smart meters, energy management systems, and government databases. While this data holds valuable information, it often exists in raw and unstructured formats, making it difficult to interpret directly. Therefore, there is a need for efficient data processing and visualization techniques that can transform raw data into meaningful insights. This project leverages the capabilities of data analytics and visualization tools to analyze electricity consumption patterns in a structured and user-friendly manner.

The dataset used in this project is collected from reliable government and verified energy-related sources, ensuring authenticity and accuracy. The analysis focuses on electricity consumption across Indian states, regions, and metropolitan cities during the years 2019 and 2020. This period is particularly significant as it captures the impact of the COVID-19 pandemic, which caused major disruptions in economic activities, industrial production, and human behavior. The lockdown measures implemented during the pandemic led to noticeable changes in electricity consumption, including reduced industrial demand and increased residential usage. Studying these changes provides valuable insights into how external factors can influence energy consumption patterns.

The primary objective of this project is to identify trends, patterns, and anomalies in electricity consumption. By analyzing data across different dimensions such as time, geography, and sector, the project aims to provide a comprehensive understanding of electricity usage. Time-based analysis includes studying daily, monthly, and seasonal variations to identify peak demand periods and low-demand intervals. Geographical analysis focuses on comparing electricity consumption across states and regions to identify high-consumption and low-consumption areas. Sector-wise analysis examines the contribution of residential, commercial, and industrial sectors to overall electricity demand.

A systematic approach is followed to achieve these objectives. The first step involves data cleaning and preprocessing using tools like Microsoft Excel and Tableau Prep. This step is crucial to ensure data quality by

removing inconsistencies, handling missing values, and standardizing formats. Clean data forms the foundation for accurate analysis and reliable results. After preprocessing, SQL is used to query and aggregate the data efficiently. SQL enables filtering, grouping, and summarizing large datasets, making it easier to extract meaningful information for analysis.

Once the data is prepared, Tableau is used to create interactive dashboards and visualizations. Tableau is a powerful data visualization tool that allows users to represent data in the form of charts, graphs, maps, and dashboards. These visualizations make it easier to understand complex datasets and identify patterns at a glance. The project includes multiple dashboards such as national-level overviews, regional comparisons, state-wise rankings, and metro city analyses. Each dashboard is designed to provide specific insights and allow users to interact with the data through filters and parameters.

One of the key features of this project is the analysis of time-based electricity consumption patterns. By examining how electricity usage varies throughout the day, month, and year, it is possible to identify peak demand periods and seasonal trends. For example, electricity consumption may increase during summer months due to the use of air conditioning, while winter months may show different patterns depending on the region. Understanding these trends helps in planning electricity generation and distribution more effectively.

Another important aspect of the project is the analysis of regional variations in electricity consumption. Different regions have different consumption patterns based on factors such as population density, level of industrialization, climate conditions, and economic activities. For instance, industrial hubs tend to have higher electricity consumption due to manufacturing activities, while rural areas may have lower consumption levels. By comparing regions, the project provides insights into disparities in electricity usage and helps identify areas that require infrastructure development or policy intervention.

The project also includes sector-wise analysis, which examines the contribution of residential, commercial, and industrial sectors to total electricity consumption. Each sector has unique usage patterns and requirements. Residential consumption is influenced by household activities and lifestyle, commercial consumption depends on business operations, and industrial consumption is driven by production processes. Understanding these differences allows energy providers to design targeted strategies for energy efficiency and demand management.

A significant highlight of this project is the comparison of electricity consumption before and after the COVID-19 lockdown. The lockdown led to a shift in electricity usage patterns, with a decline in industrial and commercial consumption and an increase in residential usage. This analysis provides valuable insights into how unexpected events can impact energy demand and highlights the importance of flexible and resilient energy systems. Such insights can help policymakers and utility companies prepare for future disruptions and ensure a stable electricity supply.

The project also emphasizes the role of data visualization in simplifying complex information. Instead of presenting data in the form of lengthy tables and reports, visualizations provide a clear and intuitive representation of data. Interactive dashboards allow users to explore data dynamically, apply filters, and focus

on specific aspects of interest. This enhances user engagement and makes it easier to derive insights without requiring advanced technical expertise.

In addition to analysis, the project highlights the importance of sustainable energy management. With increasing concerns about climate change and environmental impact, it is essential to use energy resources efficiently. By identifying high-consumption areas and peak demand periods, the project supports the development of strategies for reducing energy wastage and promoting the use of renewable energy sources. Efficient energy management not only reduces costs but also contributes to environmental sustainability.

Furthermore, this project demonstrates the practical application of academic knowledge in solving real-world problems. It integrates concepts from data analysis, database management, and visualization to create a comprehensive solution for electricity consumption analysis. Working with real datasets provides hands-on experience in handling challenges such as data inconsistency, large data volumes, and meaningful interpretation of results. This enhances technical skills as well as analytical thinking and problem-solving abilities.

The use of modern tools such as Excel, SQL, Tableau Prep, and Tableau reflects the importance of technology in data-driven decision-making. These tools enable efficient data processing, analysis, and visualization, making it possible to handle complex datasets with ease. The ability to create interactive and user-friendly dashboards ensures that the insights generated are accessible to a wide range of users, including non-technical stakeholders.

In conclusion, this project provides a comprehensive analysis of electricity consumption patterns using advanced data visualization techniques. It highlights the importance of understanding energy usage for effective planning, management, and sustainability. By transforming raw data into meaningful insights, the project supports informed decision-making and contributes to the development of efficient energy systems. It also demonstrates how the integration of data analytics and visualization can play a crucial role in addressing real-world challenges and shaping a sustainable future.

CHAPTER-2

WEEKLY OVERVIEW

CHAPTER-2

WEEKLY OVERVIEW

1st Month	Introduction to Data Analytics and Tableau Getting Started with Tableau (Installation, Interface, and Basics) Overview of electricity consumption datasets
2nd Month	Connecting to Data Sources Data Preparation and Cleaning in Tableau Ensuring accuracy for energy consumption data
3rd Month	Basic Charts and Visualizations (Bar, Line, Pie, Maps) Working with Filters and Sorting Building monthly and state-level consumption views
4th Month	Advanced Charts (Heat Maps, Scatter Plots, Tree Maps) Calculated Fields and Parameters Using parameters for Top N/Bottom N state analysis Dashboards and Stories Interactive Features (Actions, Tooltips, Drill-Downs) Ensuring consistency across dashboards
5th Month	Data Blending and Joins Connecting Tableau with Excel, SQL, and Cloud Data Ensuring the consistency across dashboards
6th Month	Project – Build an Interactive Business Dashboard using Tableau tool Story on Electrical Consumption in India Project Testing, Deployment & Documentation Final Report and GitHub Submission

CHAPTER-3

MODULES

CHAPTER-3

MODULES

Data Analytics with Tableau is the process of exploring, analyzing, and visualizing data using Tableau, a powerful business intelligence (BI) and data visualization tool. Tableau allows individuals and organizations to transform raw data into meaningful insights through interactive dashboards and visual reports. Unlike traditional reporting, Tableau emphasizes simplicity, interactivity, and storytelling with data, making insights accessible to both technical and non-technical users.

The primary goal of using Tableau is to help organizations make data-driven decisions by recognizing patterns, identifying trends, and understanding hidden relationships within datasets. Tableau is widely used in industries such as finance, healthcare, marketing, retail, and technology.

Tableau operates with a user-friendly drag-and-drop interface, making it possible to connect to multiple data sources, clean and transform data, and build dashboards without complex coding. It promotes **clarity, accuracy, and collaboration**, enabling teams to stay ahead in today's data-driven world.

The Electricity Consumption Analysis System is developed in a structured and modular manner across multiple weeks. Each phase focuses on a specific set of tasks, ensuring systematic development, easy maintenance, and scalability. This approach follows step-by-step learning and implementation, where each module builds upon the previous one to create a complete and efficient data visualization system using Tableau.

1. Data Preprocessing Module

The Data Preprocessing Module plays a crucial role in preparing the electricity consumption dataset for accurate analysis. Raw data collected from multiple sources often contains inconsistencies such as duplicate records, missing values, and irregular formats. In this module, the dataset was thoroughly examined and cleaned using tools like MS Excel and SQL. Duplicate entries were identified and removed, while missing values were handled carefully to ensure data completeness without affecting the integrity of the dataset.

In addition, field names were standardized to maintain consistency across different datasets. Data from multiple years (2019 and 2020) and different levels (state-wise, region-wise, and metro city data) were merged into a single unified dataset. This ensured that comparisons could be performed easily across time and geography. Data types such as dates and numerical values were also verified and corrected wherever necessary.

Overall, this module ensures that the dataset is clean, structured, and reliable. Proper preprocessing reduces errors in analysis and improves the accuracy of visualizations created in Tableau. It forms the foundation of the entire project, as all further analysis depends on the quality of the data.

2. Feature Analysis Module

The Feature Analysis Module focuses on identifying and analyzing key attributes that influence electricity consumption. Important features such as state-wise consumption, region-wise usage, metro city

demand, and time-based variations (monthly, quarterly, yearly) were studied in detail. This analysis helps in understanding how electricity usage differs across various dimensions.

Using Tableau, calculated fields were created to derive additional insights from the data. These included percentage growth in consumption, seasonal peaks, and comparisons between pre-lockdown and post-lockdown periods. Such calculations enhanced the analytical depth of the project and provided meaningful interpretations of the data.

This module plays an important role in identifying patterns and trends in electricity consumption. It highlights the factors that influence demand and provides a deeper understanding of how electricity usage changes over time and across regions. These insights are valuable for decision-making and future planning.

3. Visualization Module

The Visualization Module is responsible for transforming processed data into graphical representations using Tableau. Various types of charts and graphs were created to represent electricity consumption patterns effectively. Bar charts were used to compare state-wise consumption, line charts were used to analyze monthly and quarterly trends, pie charts were used for proportional analysis, and maps were used to display geographical distribution.

In addition to basic charts, KPI cards were created to display key metrics such as total electricity consumption, highest consuming state, and growth trends. These visualizations made it easier to understand complex data at a glance. The use of colors, labels, and formatting improved the clarity and readability of the dashboards.

Storylines were also developed to present a complete narrative of electricity consumption in India. Special focus was given to showing the impact of COVID-19 by comparing data before and after lockdown periods. This module ensures that data is presented in a clear, engaging, and meaningful way.

4. Interactive Dashboard Module

The Interactive Dashboard Module combines multiple visualizations into a single, user-friendly interface. Dashboards allow users to analyze data from different perspectives without switching between multiple sheets. In this project, dashboards were designed to provide a comprehensive view of electricity consumption across states, regions, and metro cities.

Interactive features such as filters and parameters were added to enhance user experience. Filters allowed users to select specific years, states, or regions, while parameters enabled dynamic analysis such as Top N and Bottom N states. Drill-down features were also included, allowing users to explore data in greater detail.

This module plays a key role in making the project interactive and user-centric. It provides stakeholders with an intuitive platform to explore data, identify trends, and make informed decisions. The dashboards are designed to be simple, responsive, and easy to understand.

5. Web Integration Module

The Web Integration Module focuses on deploying the Tableau dashboards in a web environment. In this project, Tableau dashboards were integrated with Flask to make them accessible through a web interface. This allows users to view and interact with dashboards without requiring Tableau Desktop.

This integration ensures real-time interaction and improves accessibility for a wider audience, including policymakers, analysts, and researchers. Users can access the dashboards from anywhere, making it easier to share insights and collaborate.

This module demonstrates the practical implementation and deployment of the project. It highlights how data visualization solutions can be extended beyond development environments and used in real-world applications for decision-making.

Tools & Techniques Used

1. Tableau Desktop

Tableau Desktop is the primary tool used for creating interactive dashboards and visualizations in this project. It provides a user-friendly interface that allows users to create charts and graphs using simple drag-and-drop functionality. Various features such as calculated fields, parameters, filters, and trend lines were used to enhance analysis.

Tableau was used to build visual stories that showcase electricity consumption patterns across different states, regions, and time periods. It enabled the transformation of raw data into meaningful insights that are easy to understand. The interactive nature of Tableau dashboards allowed users to explore data dynamically. Overall, Tableau played a central role in the project by providing powerful visualization capabilities and supporting data-driven decision-making**.**

2. SQL (MySQL Workbench)

SQL was used as a supporting tool for data validation and preprocessing. Queries were written in MySQL Workbench to check for missing values, duplicates, and inconsistencies in the dataset. SQL helped in performing operations such as filtering, grouping, and aggregating data efficiently.

It was also used to merge datasets from different years (2019 and 2020), ensuring a unified dataset for analysis. SQL provided quick insights into electricity consumption data and helped in verifying the accuracy of the dataset before importing it into Tableau.The use of SQL improved data reliability and ensured that the dataset used for visualization was accurate and consistent.

3. MS Excel / CSV

MS Excel was used as an intermediate tool for organizing and formatting the dataset. It helped in performing initial preprocessing tasks such as arranging columns, handling null values, and standardizing data formats. Excel made it easy to view and edit data before importing it into Tableau.

CSV files were used for storing and transferring data between tools. The cleaned dataset was saved in CSV format and imported into Tableau for further analysis. Excel also helped in verifying data accuracy and making quick corrections when needed.Overall, Excel served as a simple and effective tool for preparing data, ensuring smooth integration with Tableau, and supporting the overall workflow of the project.

CHAPTER-4

PROJECT

CHAPTER-4

PROJECT

Overview:

"Plugging into the Future: An Exploration of Electricity Consumption Patterns Using Tableau" is a data visualization–based project designed to analyze electricity consumption across different regions, states, and metro cities in India. The primary goal of this system is to transform raw electricity usage data into meaningful insights using advanced data visualization techniques, reducing manual analysis and improving decision-making efficiency.

This project utilizes Tableau, a powerful data visualization tool, to explore electricity consumption trends based on time, geography, and sector. The analysis focuses on identifying peak demand periods, seasonal variations, and the impact of socio-economic factors such as the COVID-19 lockdown. By leveraging interactive dashboards, the system enables users to easily understand complex datasets and derive actionable insights.

The project covers all stages of a complete data analytics pipeline, including data acquisition, preprocessing, analysis, visualization, and deployment. The resulting system demonstrates how data visualization can support energy planning, policy-making, and sustainable electricity management.

Objectives

• Analyze electricity consumption across states, regions, and metro cities
• Identify trends such as peak demand, seasonal variations, and anomalies
• Compare electricity usage before and after COVID-19 lockdown
• Build interactive dashboards using Tableau for easy data exploration
• Support decision-making for energy providers and policymakers
• Demonstrate practical application of data analytics and visualization

System Architecture

The system follows a structured data analytics pipeline consisting of the following layers:

1. **Data Acquisition Layer** – Collects electricity consumption datasets from government and verified sources
2. **Preprocessing Layer** – Cleans and standardizes data using Excel and SQL
3. **Analysis Layer** – Performs aggregation, filtering, and trend analysis
4. **Visualization Layer** – Creates charts, graphs, and dashboards using Tableau
5. **Deployment Layer** – Integrates dashboards with Flask for web-based access

Data Flow

Dataset (Excel/CSV) → Data Cleaning (Excel/SQL) → Tableau Analysis → Dashboard Creation → Web Integration (Flask)

Development Process

The project was developed in structured phases to ensure clarity and efficiency.

1.Data Acquisition

• Electricity consumption data was collected from reliable sources
• Data included state-wise, region-wise, and metro-level consumption
• Multiple datasets (2019 and 2020) were combined for comparison

2.Data Preprocessing

• Data was cleaned using Excel and SQL
• Missing values and duplicates were removed
• Data formats were standardized for consistency

3.Data Analysis

• SQL queries were used for aggregation and filtering
• Trends were analyzed across time and regions
• Key metrics such as growth rate and peak demand were identified

4.Visualization

• Tableau was used to create charts and dashboards
• Bar charts, line charts, maps, and KPI cards were developed
• Interactive filters and parameters were added

5.Deployment

• Dashboards were integrated using Flask
• Users can access dashboards via web interface
• Improved accessibility and usability

Key Features

1. **Interactive Dashboards**
   Allows users to explore electricity consumption using filters and parameters
2. **Comparative Analysis**
   Enables comparison between 2019 and 2020 (COVID impact)

3. **Geographical Insights**

   Visualizes state-wise and region-wise electricity usage

4. **Time-Series Analysis**

   Identifies monthly, quarterly, and seasonal trends

5. **Scalability**

   Can be extended with real-time or additional datasets

6. **User-Friendly Interface**

   Easy-to-understand visualizations for non-technical users

Challenges and Solutions

• **Challenge:** Missing and inconsistent data

Solution: Data cleaning using Excel and SQL

• **Challenge:** Large dataset handling

Solution: Efficient aggregation and filtering

• **Challenge:** Complex data interpretation

Solution: Use of visual dashboards and storytelling

Results and Outcomes

The project successfully generated meaningful insights into electricity consumption patterns.

• Identified high-consumption states and regions

• Observed seasonal peaks in electricity demand

• Detected significant reduction during COVID-19 lockdown

• Developed interactive dashboards for analysis

• Demonstrated the effectiveness of Tableau in data visualization

Final Outcomes

• Fully functional Tableau-based dashboard system

• Successful analysis of electricity consumption data

• Interactive and user-friendly visualization platform

• Practical implementation of data analytics concepts

Future Scope

• Integration of real-time electricity data

• Use of machine learning for demand forecasting

• Cloud deployment for wider accessibility

• Mobile-friendly dashboard applications

• API integration for live data updates

System Description

1. Data Source – Excel Dataset

The Excel sheet represents the raw dataset used in the project. It contains structured electricity consumption data with the following fields:

- **States** – Name of Indian states
- **Regions** – Categorized as NR, SR, WR, ER, NER
- **Latitude & Longitude** – Used for geographical mapping
- **Dates** – Time-based data for analysis
- **Usage** – Electricity consumption values

A	B	C	D	E	F	G	H	I
States	Regions	latitude	longitude	Dates	Usage			
Punjab	NR	31.51997	75.98	02-01-2019	119.9			
Haryana	NR	28.45001	77.01999	02-01-2019	130.3			
Rajasthan	NR	26.45	74.63998	02-01-2019	234.1			
Delhi	NR	28.66999	77.23	02-01-2019	85.8			
UP	NR	27.59998	78.05001	02-01-2019	313.9			
Uttarakhai	NR	30.32041	78.05001	02-01-2019	40.7			
HP	NR	31.10003	77.1666	02-01-2019	30			
J&K	NR	33.45	76.24	02-01-2019	52.5			
Chandigarl	NR	30.72	76.78001	02-01-2019	5			
Chhattisga	WR	22.09042	82.15999	02-01-2019	78.7			
Gujarat	WR	22.2587	71.1924	02-01-2019	319.5			
MP	WR	21.30039	76.13002	02-01-2019	253			
Maharasht	WR	19.25023	73.16017	02-01-2019	428.6			
Goa	WR	15.492	73.818	02-01-2019	12.8			
DNH	WR	20.26658	73.01662	02-01-2019	18.6			
Andhra Pra	SR	14.75043	78.57003	02-01-2019	164.6			
Telangana	SR	18.1124	79.0193	02-01-2019	204.2			
Karnataka	SR	12.57038	76.92	02-01-2019	206.3			
Kerala	SR	8.900373	76.56999	02-01-2019	72.7			
Tamil Nad	SR	12.92039	79.15004	02-01-2019	268.3			
Pondy	SR	11.93499	79.83	02-01-2019	6.3			
Bihar	ER	25.78541	87.47997	02-01-2019	82.3			
Jharkhand	ER	23.80039	86.41999	02-01-2019	24.8			
Odisha	ER	19.82043	85.90002	02-01-2019	70.2			
West Beng	ER	22.58039	88.32995	02-01-2019	108.2			
Sikkim	ER	27.33333	88.61665	02-01-2019	2			

Consumption

Fig 4.1: DataSet

This dataset serves as the foundation of the project. It is manually cleaned and formatted to remove inconsistencies such as missing values and incorrect entries. The structured format allows easy integration with Tableau for visualization.

2. Backend Processing – SQL (MySQL Workbench)

The MySQL Workbench screen shows how the dataset is managed and queried using SQL.

Key operations performed:

- **Creation of database:**

  create schema electricity;

- **Selecting dataset:**

  select * from consumption;

- **Extracting important fields:**

  select States, Regions from consumption;

- **Sorting data by date:**

  select * from consumption order by dates;

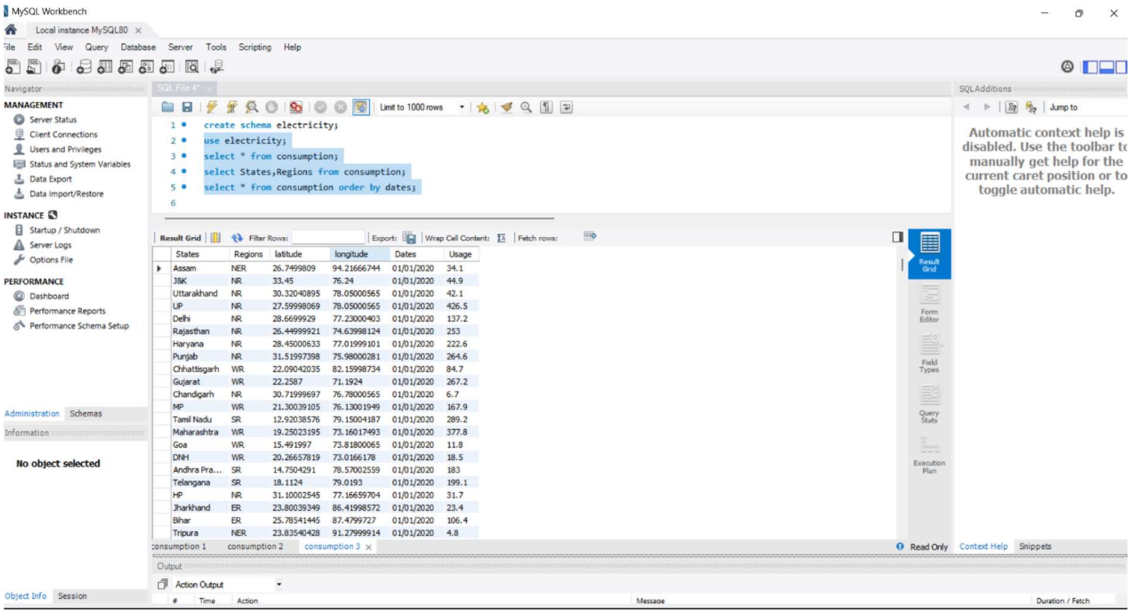

Fig 4.2: Data Preprocessing Using SQL

The result grid displays organized data with columns like state, region, coordinates, date, and usage. SQL ensures:

- Data accuracy
- Efficient querying
- Structured data retrieval
- Easy integration with Tableau

Model Deployment and Integration

To transform the electricity consumption analysis project into an interactive application, a **Flask-based web interface** was developed. This integration allows users to access electricity consumption dashboards through a browser, enabling real-time interaction with data visualizations created in Tableau. Users can explore electricity usage patterns, compare regions, and analyze trends dynamically without requiring Tableau Desktop.

Flask, a lightweight Python web framework, is used to connect the backend and frontend of the system. It allows seamless integration between Tableau dashboards and a web interface. The backend handles user requests, loads dashboards, and renders web pages dynamically. This approach makes the project more accessible and demonstrates its real-world applicability for policymakers, analysts, and researchers.

1.Flask Framework Overview

Flask was chosen for deployment due to its simplicity, flexibility, and ease of integration with visualization tools. It provides a minimal yet powerful framework for building web applications. In this project, Flask acts as a bridge between Tableau dashboards and the user interface.

The backend handles:

- Rendering web pages for dashboard access
- Managing navigation between pages
- Embedding Tableau dashboards into the web interface
- Handling user interactions through requests

2.Backend Implementation

```
from flask import Flask, render_template

# Initialize Flask App

app = Flask(__name__)

# Home Page Route

@app.route('/')

def home():

    return render_template('index.html')

# Dashboard Page Route

@app.route('/dashboard')

def dashboard():

    return render_template('dashboard.html')
```

```
# Run Application
if __name__ == '__main__':
    app.run(debug=True)
```

3. Front-End Templates

The front-end of the Electricity Consumption project is developed using **HTML, CSS, and Javascript**, providing a modern, responsive, and user-friendly interface. The design is based on a professional Bootstrap template and customized to display electricity consumption insights, dashboards, and analysis.

The interface includes multiple sections such as Home, About, Dashboard, Story, Visualizations, and Conclusion. These sections allow users to navigate easily and explore electricity consumption data interactively.

index.html – Main Web Page

The main page serves as the entry point of the application. It includes navigation menus, hero section, project overview, dashboards, and insights.

Below is the integrated template used in the project:

```
<!-- Electricity Consumption Main UI -->
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Electricity Consumption Analysis</title>
  <!-- Bootstrap & CSS -->
  <linkhref="{{url_for('static',filename='assets/vendor/bootstrap/css/bootstrap.min.css') }}" rel="stylesheet">
  <link href="{{ url_for('static', filename='assets/css/main.css') }}" rel="stylesheet">
</head>
<body>
<!-- Header -->
<header>
  <h1>Electricity Consumption Analysis</h1>
  <nav>
    <a href="#hero">Home</a>
```

```
    <a href="#about">About</a>
    <a href="#services">Dashboard</a>
    <a href="#portfolio">Story</a>
    <a href="#visualizations">Visualizations</a>
    <a href="#conclusion">Conclusion</a>
  </nav>
</header>
<!-- Hero Section -->
<section id="hero">
  <h2>Plugging into the Future</h2>
  <p>Explore electricity consumption patterns across India</p>
</section>
<!-- About Section -->
<section id="about">
  <h2>About Project</h2>
  <p>
    This project analyzes electricity consumption across Indian states
    to understand trends, regional differences, and energy demand.
  </p>
</section>
<!-- Tableau Dashboard Section -->
<section id="services">
  <h2>Dashboard</h2>
  <!-- Tableau Embed -->
  <div class='tableauPlaceholder'>
    <object class='tableauViz'>
      <param name='host_url' value='https://public.tableau.com/' />
      <param name='name' value='proj1/Dashboard1' />
```

```
    </object>
  </div>
</section>
<!-- Visualizations Section -->
<section id="visualizations">
  <h2>Analysis</h2>
  <p>State-wise, Region-wise, and Time-based electricity analysis</p>
</section>
<!-- Conclusion -->
<section id="conclusion">
  <h2>Conclusion</h2>
  <ul>
    <li>Maharashtra is highest consumer</li>
    <li>Gujarat is second highest</li>
    <li>Sikkim is lowest consumer</li>
  </ul>
</section>
</body>
</html>
```

4. Deployment Workflow

1. **Data Preparation:**
   Electricity consumption data is cleaned and processed using Excel and SQL.
2. **Visualization Creation:**
   Interactive dashboards are built in Tableau using charts, maps, and KPIs.
3. **Dashboard Publishing:**
   Dashboards are published to Tableau Public or Server.
4. **Flask Integration:**
   Tableau dashboards are embedded into a Flask web application.
5. **User Interaction:**
   Users access the web page, explore dashboards, and interact using filters and parameters.

5. Role of Deployment in This Project

Makes dashboards accessible via web browser

• Enables real-time interaction with electricity data

• Improves usability for non-technical users

• Demonstrates real-world implementation of data analytic

This project validates the feasibility of deploying a data visualization–based electricity consumption analysis system as a real-world application. By combining Tableau dashboards with Flask integration, the system provides an interactive and accessible platform for analyzing electricity usage patterns.

The results highlight how data visualization can support efficient energy planning, policy-making, and sustainable resource management. The project demonstrates the power of combining data analytics tools with web technologies to create practical and user-friendly solutions.

The project successfully demonstrates how electricity consumption data can be transformed into meaningful insights and deployed as an interactive web application. By integrating Tableau with Flask, the system provides a scalable and user-friendly platform that can be used by policymakers, researchers, and energy providers.

It highlights the importance of data-driven decision-making and showcases how modern technologies can contribute to efficient energy management and sustainable development.

CHAPTER-5

RESULTS AND CONCLUSIONS

CHAPTER-5

RESULTS AND CONCLUSIONS

The Electricity Consumption Analysis Project demonstrates how data analytics and visualization tools can transform raw electricity usage data into meaningful insights. By using Tableau dashboards and web integration, the system provides an interactive and user-friendly platform for analyzing electricity consumption across different regions of India. After systematic development and analysis, the project successfully highlights trends, patterns, and variations in electricity usage, supporting better decision-making in the energy sector.

1. Accurate Analysis of Electricity Consumption

At the core of this project lies the ability to analyze electricity consumption data across states, regions, and metro cities with high accuracy. The dataset includes electricity usage data for the years 2019 and 2020, allowing comparative analysis. The data undergoes preprocessing steps such as cleaning, removal of duplicates, and standardization before being visualized in Tableau.

The dashboards effectively represent electricity usage patterns using bar charts, line graphs, and maps. These visualizations help identify high and low consumption regions. For example, Maharashtra is identified as the highest electricity-consuming state, while Sikkim shows the lowest consumption. Time-based analysis also highlights seasonal peaks and monthly trends in electricity usage.

This accurate analysis helps stakeholders understand electricity demand and make informed decisions. It reduces manual analysis effort and ensures reliable interpretation of complex datasets.

2. Interactive and Informative Visualization

Unlike traditional static reports, this project provides an interactive visualization experience using Tableau dashboards. Users can explore electricity consumption data dynamically through filters, parameters, and drill-down features.

The dashboards allow users to analyze data based on different dimensions such as state, region, year, and time period. For example, users can compare electricity usage before and after the COVID-19 lockdown or identify top and bottom consuming states using parameter-based controls.

This interactive approach enhances user engagement and makes data analysis more intuitive. It transforms complex numerical data into easy-to-understand visual insights, making it useful for both technical and non-technical users.

3. Seamless User Experience

The project includes a web-based interface developed using Flask, HTML, CSS, and Bootstrap. This interface allows users to access Tableau dashboards through a browser without requiring Tableau Desktop. The front-end design is clean, responsive, and easy to navigate. It includes sections such as Home, About, Dashboard, Story, Visualizations, and Conclusion. Users can easily explore different aspects of electricity consumption analysis through a structured layout.

The integration between Flask and Tableau ensures smooth interaction and real-time visualization. This enhances accessibility and makes the system suitable for wider use, including policymakers, researchers, and students.

4. Data Accuracy and Reliability

Ensuring data accuracy is a critical aspect of this project. The dataset is carefully cleaned and validated using tools like Excel and SQL before being used in Tableau. Missing values, duplicates, and inconsistencies are handled to maintain data integrity.

Proper aggregation and validation techniques are applied to ensure reliable results. The dashboards are designed to reflect accurate insights, avoiding misleading interpretations. This ensures that the analysis is trustworthy and can be used for decision-making.Maintaining data accuracy improves the credibility of the project and enhances its practical usefulness in real-world applications.

5. Scalability and Future Expansion

The project is designed with scalability in mind, allowing future enhancements and extensions. New datasets can be easily added to analyze electricity consumption for additional years or regions.

The system can also be extended to include real-time data integration, predictive analytics, and machine learning models for forecasting electricity demand. Integration with cloud platforms and APIs can further enhance the system's capabilities.

This flexibility ensures that the project can evolve into a more advanced energy analytics platform in the future.

6. Key Features of the System

• Interactive Tableau dashboards for dynamic data exploration

• Comparative analysis of electricity consumption (2019 vs 2020)

• State-wise, region-wise, and metro-level analysis

• Time-based analysis (monthly, quarterly, seasonal trends)

• COVID-19 impact analysis on electricity demand

• Web-based access using Flask integration

7. Practical Outcomes and Insights

The project successfully generated valuable insights into electricity consumption patterns in India:

• Maharashtra identified as the highest electricity-consuming state

• Gujarat as the second highest consumer

• Sikkim as the lowest electricity-consuming state

• Significant drop in electricity demand during COVID-19 lockdown

• Seasonal peaks observed during specific months

These insights demonstrate the importance of data-driven analysis in energy management and policy planning.

Web Interface Description

The web interface of the Electricity Consumption Analysis project includes multiple sections, each designed to provide an intuitive and efficient user experience. The interface allows users to explore electricity consumption data through interactive dashboards and visualizations.

1. Home Page

The Home Page acts as the entry point of the application. It displays the project title **"Electricity Consumption"** along with a navigation bar that includes sections such as Home, About, Dashboard, Story, Visualizations, Conclusion, and Blog. This structured navigation allows users to move easily between different parts of the application.

The main heading, **"Plugging into the Future: An Exploration of Electricity Consumption Patterns"**, introduces the theme of the project. It is supported by a brief description explaining the importance of electricity consumption in India, including its role in economic growth and infrastructure development. A "Get Started" button is provided to guide users toward exploring the dashboard and analysis sections.

This section provides an overview of the project and creates an engaging first impression by combining clear design with meaningful information.

Fig 5.1: Home Screen

2. About Section

The About section provides a detailed explanation of the project's objective. It focuses on analyzing electricity consumption patterns across different states in India to understand variations in regional demand, usage trends, and energy distribution.

This section highlights important functionalities such as state-wise comparison of electricity consumption, identification of the highest and lowest consuming states, and the use of interactive Tableau dashboards for visualization. It also emphasizes how the project helps in understanding energy usage patterns and supports better planning for sustainable energy management.

The content in this section helps users gain a deeper understanding of the project's scope and its real-world significance

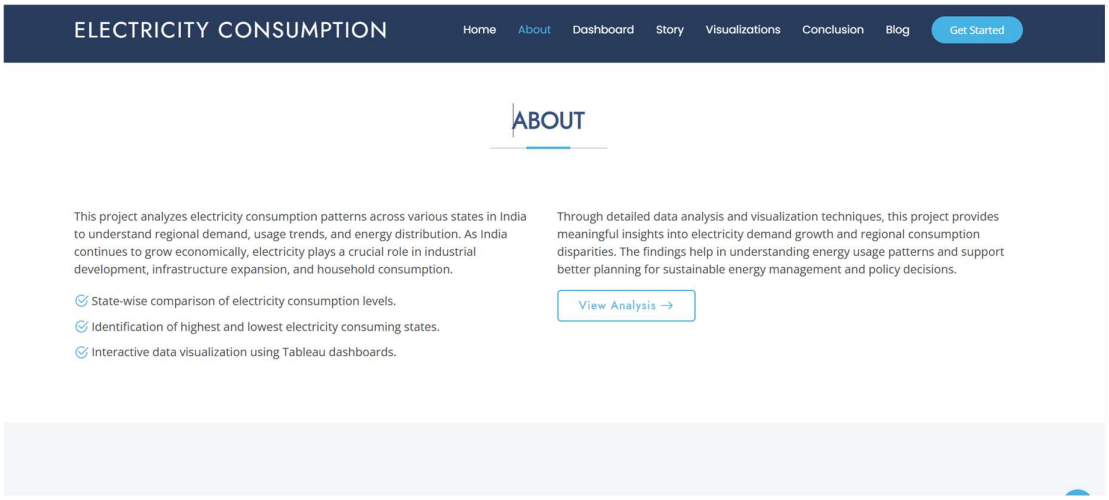

Fig 5.2: About Section

3. Dashboard Section

The Dashboard section contains interactive Tableau visualizations that present electricity consumption data in a graphical format. It includes state-wise consumption maps for the years 2019 and 2020, which allow users to compare electricity usage across different states.

In addition, region-wise pie charts display the distribution of electricity consumption among different regions, while total consumption bar graphs provide a clear comparison of usage levels. These visualizations make it easier to interpret complex data and identify patterns.

Users can interact with the dashboards using filters and legends, enabling dynamic exploration of electricity consumption data across different dimensions.

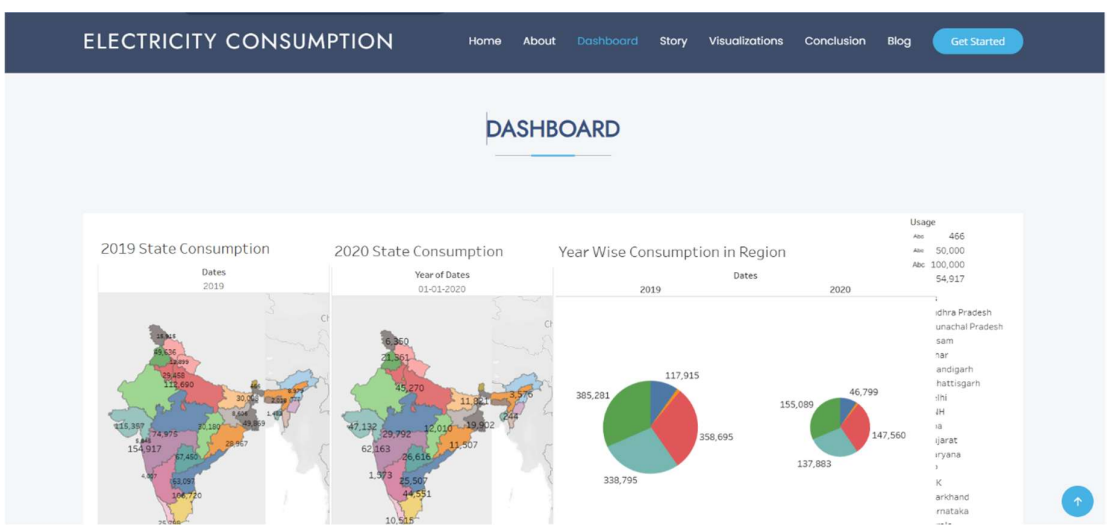

Fig 5.3: Dashboard Screen1

4. Advanced Dashboard Analysis

This section provides additional dashboards for deeper and more detailed analysis. It includes visualizations such as Top N states, which highlight the highest electricity-consuming states, and Bottom N states, which identify regions with the lowest consumption.

Comparative analysis between 2019 and 2020 is also included to show changes in electricity usage over time. Buttons such as “Dashboard 1”, “Dashboard 2”, and “Dashboard 3” allow users to switch between different views and explore multiple analytical perspectives.

This section enhances the analytical capability of the system by providing more focused and detailed insights.

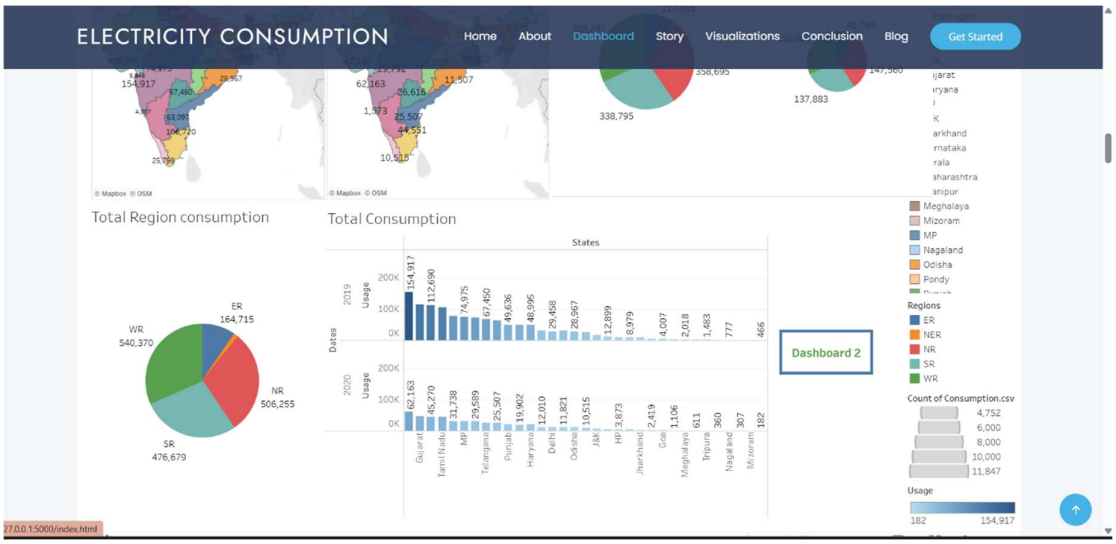

Fig 5.4: Dashboard Screen 2

Fig 5.5: Dashboard Screen3

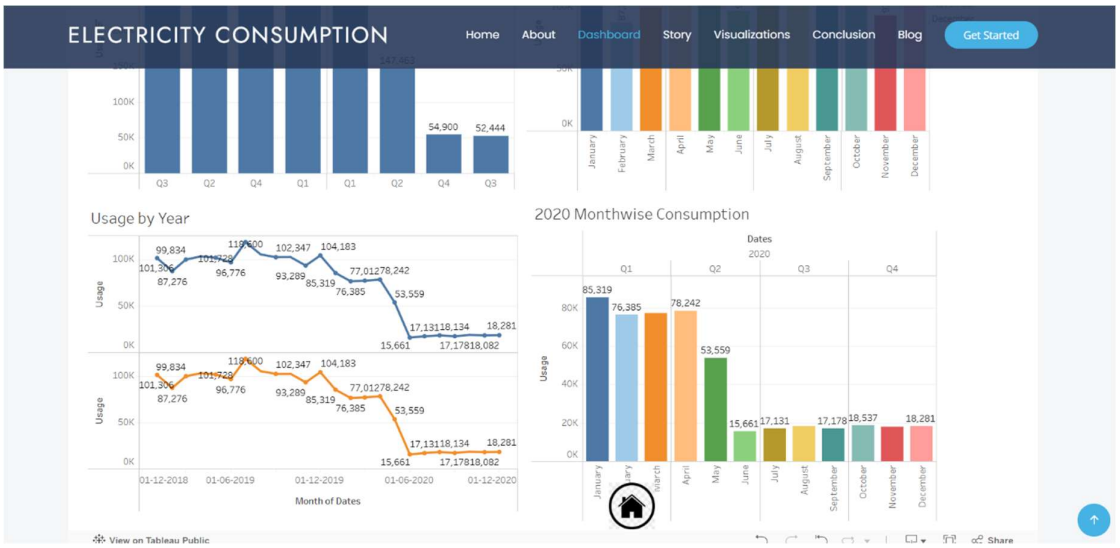

Fig 5.6: Dashboard Screen4

5. Story Section

The Story section presents electricity consumption data in a structured and sequential format using Tableau's storytelling feature. It includes multiple story points such as state-wise consumption for different years, regional comparisons, and identification of top and bottom consuming states.

It also covers time-based analysis, including monthly and yearly trends. By presenting data step-by-step, this section helps users understand the analysis in a logical and easy-to-follow manner.

The storytelling approach improves clarity and makes complex data more understandable.

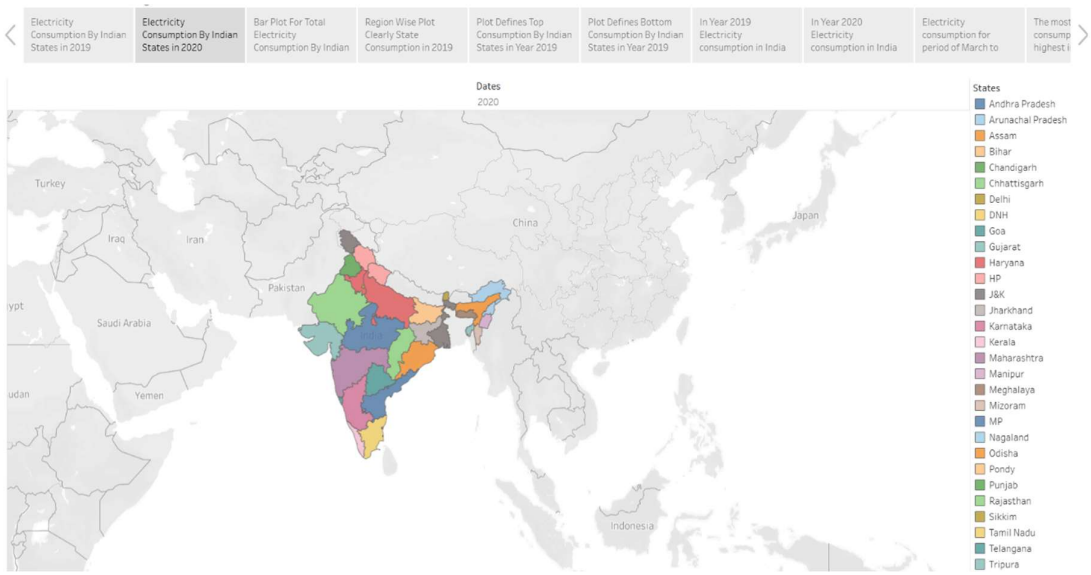

Fig 5.7: Story Screen1

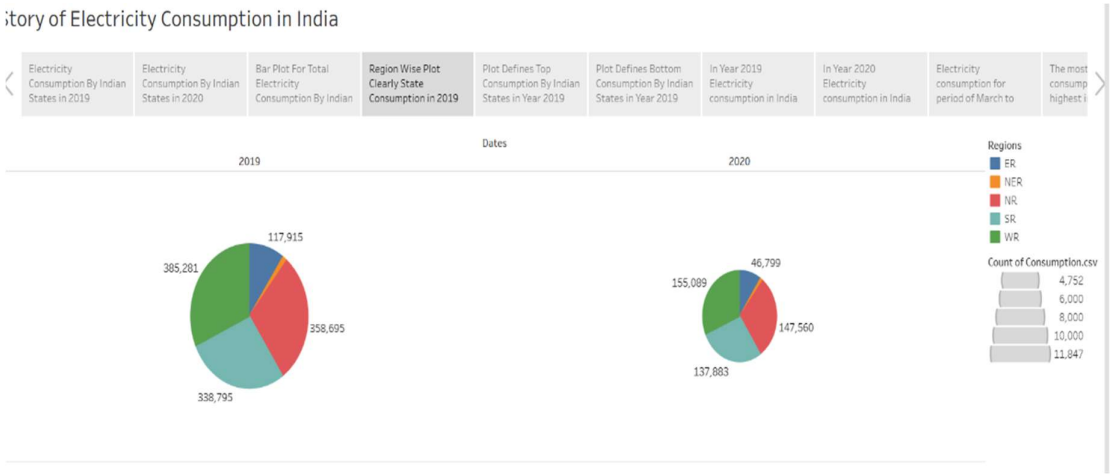

Fig 5.8: Story Screen2

6. Visualizations Section

The Visualizations section provides detailed graphical representations of electricity consumption data. It includes line charts that show year-wise trends, bar charts that represent monthly consumption, and time-series graphs that highlight seasonal variations.

These visualizations help in identifying important patterns such as fluctuations in electricity demand over time. For example, the charts show a noticeable drop in consumption during the COVID-19 lockdown period and increased demand during summer months.

The graphical representation of data allows users to easily interpret trends and variations.

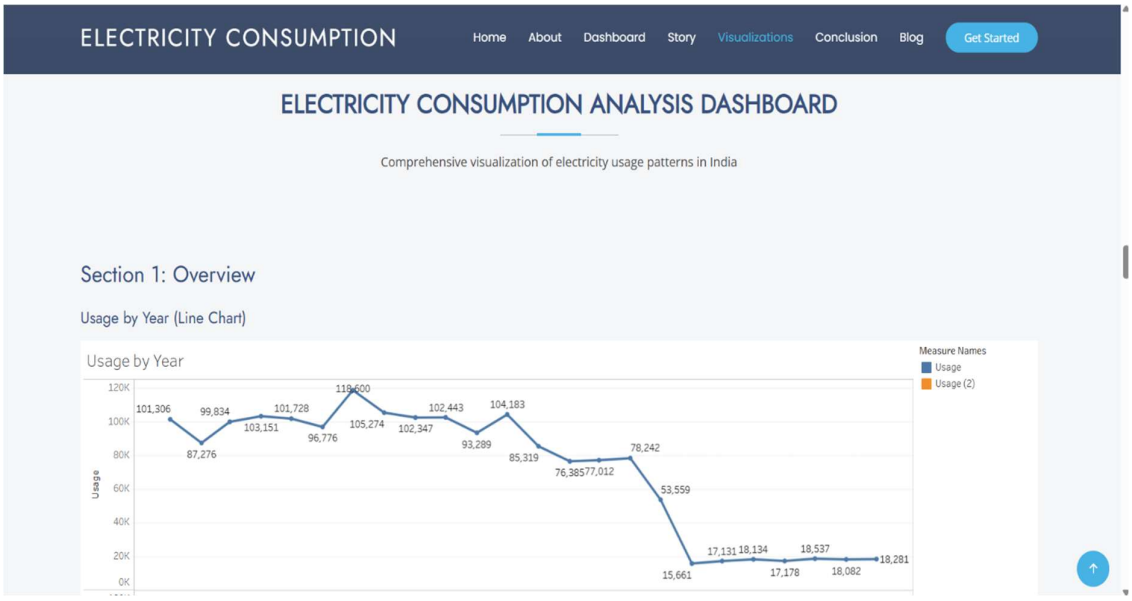

Fig 5.9: Visualization Screen1

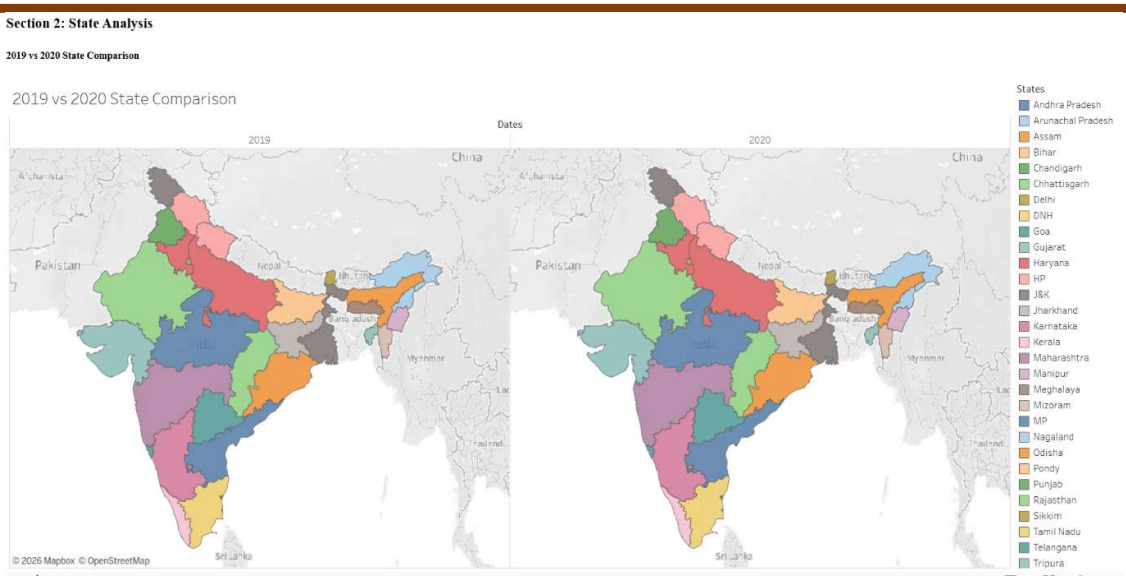

Fig 5.10: Visualization Screen2

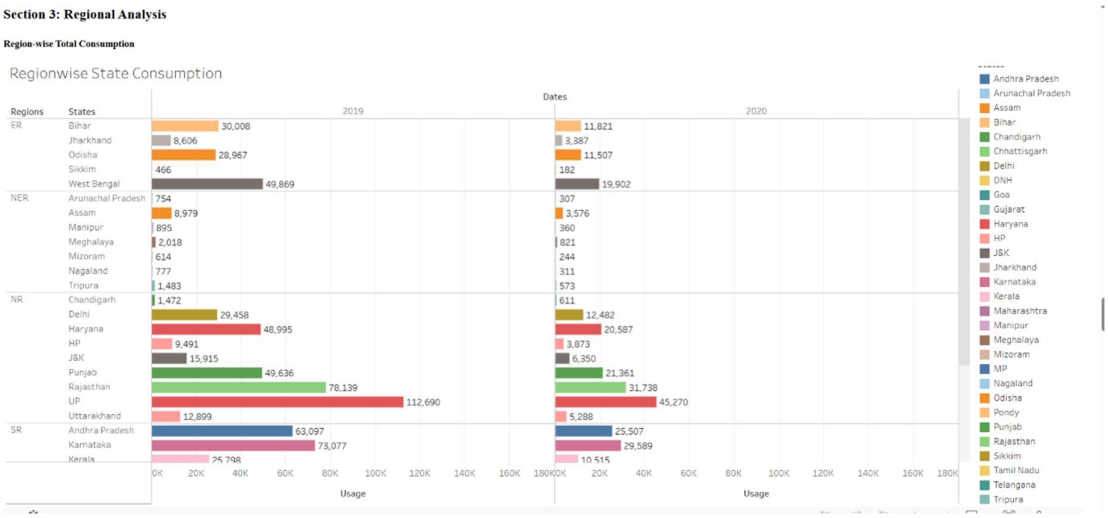

Fig 5.11: Visualization Screen3

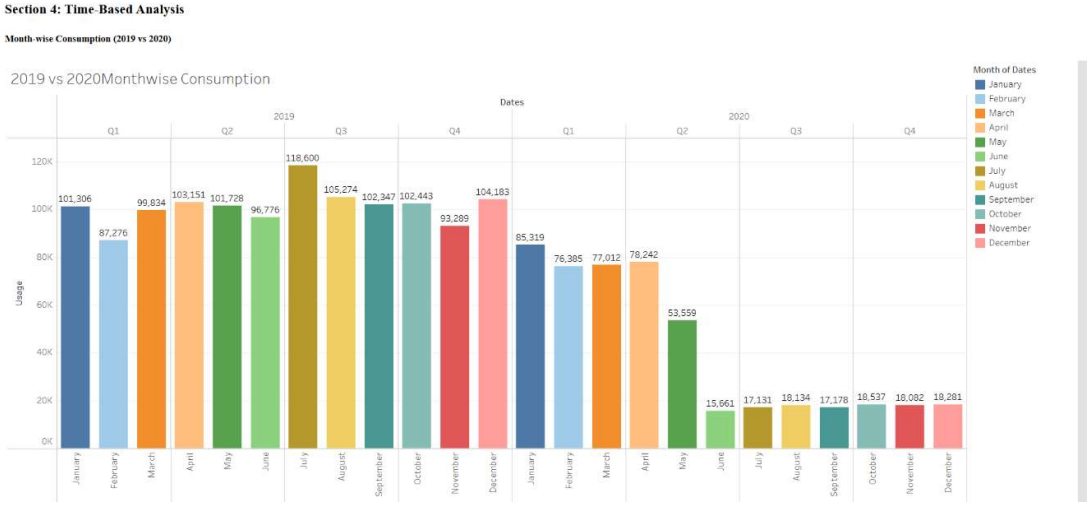

Fig 5.12: Visualization Screen4

7. Conclusion Section

The Conclusion section presents the final summary of the project. It highlights the major observations derived from the analysis, including consumption trends across different states, regional differences, and the influence of external factors such as the COVID-19 pandemic.

It consolidates all findings into a clear and concise summary, helping users understand the overall outcome of the project. This section effectively wraps up the analysis and reinforces the importance of data-driven decision-making in energy management.

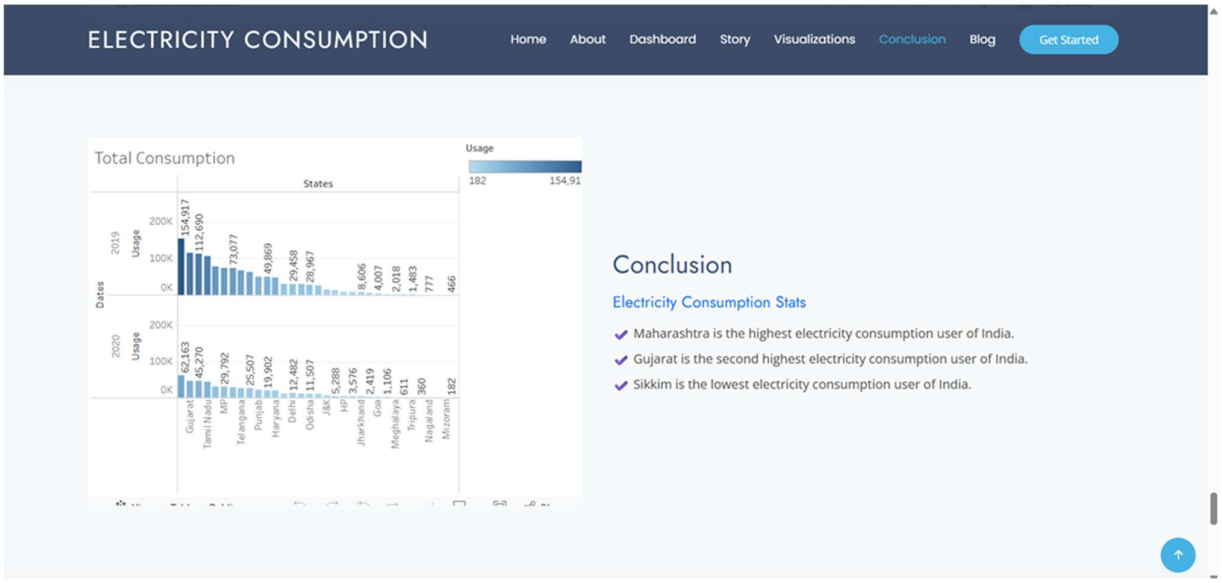

Fig 5.13: Conclusion Screen

8. Project Insights Section

This section summarizes key findings derived from the analysis. It highlights that Maharashtra is the highest electricity-consuming state, followed by Gujarat, while Sikkim has the lowest consumption.

It also presents insights related to the impact of COVID-19, showing a reduction in electricity demand due to decreased industrial and commercial activities. Additionally, it identifies that Southern and Western regions generally have higher electricity usage compared to other regions.

These insights provide a clear understanding of the overall trends and patterns observed in the data.

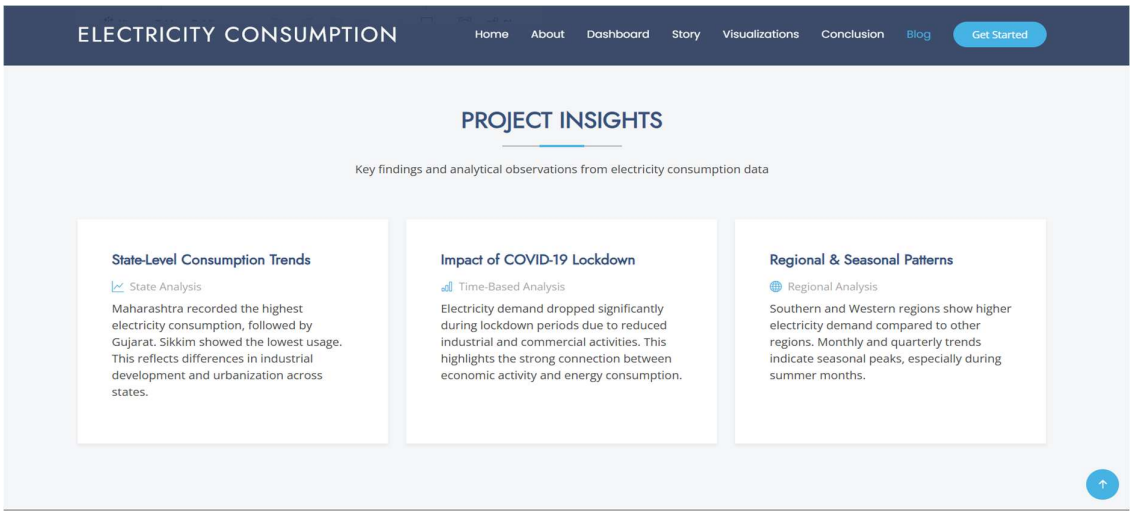

Fig 5.14: Project Insights

The web interface successfully integrates data visualization and web technologies to provide a complete platform for analyzing electricity consumption. It enables users to explore data interactively, understand patterns through visualizations, and gain meaningful insights into electricity usage across India.

Conclusion

The Electricity Consumption Analysis project demonstrates the effective application of data analytics and visualization techniques to understand electricity usage patterns across India. By combining powerful tools like Tableau for visualization and Flask for web integration, the system provides an interactive and accessible platform for analyzing complex electricity consumption data.

The project efficiently analyzes electricity usage across states, regions, and time periods, enabling the identification of key trends such as peak demand, seasonal variations, and regional disparities. The comparative analysis between 2019 and 2020 clearly highlights the impact of external factors such as the COVID-19 lockdown, proving the importance of time-based data analysis in understanding real-world scenarios.

The use of data preprocessing techniques and structured dataset management ensures accuracy and reliability in the analysis. Interactive dashboards enhance user experience by allowing dynamic exploration through filters, parameters, and drill-down features. This makes the system easy to use for both technical and non-technical users, including policymakers, researchers, and students.

Overall, the project successfully demonstrates how data visualization can transform raw electricity data into meaningful insights. It highlights the role of data-driven decision-making in energy planning, efficient resource utilization, and sustainable development. With further enhancements such as real-time data integration and predictive analytics, this system can evolve into a comprehensive energy analytics platform for future applications.

CHAPTER-6

VERIFIABLE CREDENTIALS

CHAPTER-6

VERIFIABLE CREDENTIALS

Verifiable Link for Certificate :

https://apsche.smartinternz.com/certificate/student/c07381f59ad54420a15e3ffc688a6668

GitHub Repository Link :

https://github.com/kaipunarasimha74/Electricity-Consumption-Analysis

CODE QR :